

HAMOUDA CHEKIR

Software Engineer, Full-Stack and Applied Artificial Intelligence

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PROFILE

Software engineer, ESPRIT graduate (TWIN specialisation), focused on delivery. I design and ship reliable web products using full-stack development, applied AI and microservices architectures. Most recent work at Talan: NextHire, an AI-first applicant tracking system of 16 microservices (React, Node.js, Python, FastAPI, MongoDB, Docker).

PROFESSIONAL EXPERIENCE

Software Engineer, Final-Year Project

February 2026 - August 2026

Talan, Tunis, Tunisia

- Delivered an AI-first applicant tracking system (NextHire) of 16 microservices: 1 Node.js/Express gateway, 2 React SPAs and 13 Python FastAPI/Flask services, on a shared MongoDB architecture with Dockerised service boundaries.
- Automated the interview flow with an adaptive LLM voice agent and a YOLO integrity module (face verification, prohibited-object detection) running on live WebRTC sessions.
- Implemented a LangGraph pipeline fusing transcription, behavioural and visual signals into scored, ranked hiring reports; automated CV parsing (spaCy, PaddleOCR) with anti-hallucination checks.

Full-Stack Developer Intern

June 2025 - August 2025

SmartConseil / Dafe Management, Tunis, Tunisia

- Delivered a modular Laravel back office (regions, packs, modules) by rebuilding CRUD workflows in Blade and JavaScript on PostgreSQL and MongoDB.
- Centralised fragmented crawler tooling (HTML and API drivers, grids, IP modules) into a single codebase; shipped in Agile with code reviews, unit tests, SonarQube and GitLab CI/CD.

Web Developer Intern

June 2024 - August 2024

Big Deal (Club Privileges), Tunis, Tunisia

- Shipped a digital gift-code module (generation, assignment, tracking) on Laravel and Filament, enabling end-to-end promotional campaigns.

Web Developer Intern

February 2023 - May 2023

HOTIX, Tunis, Tunisia

- Launched the Consommi Tounsi #619 e-commerce site: core storefront features in .NET, Scrum delivery, full test pass and production deployment.

ACADEMIC PROJECTS

TalentMatch, a secure auction platform (AI and blockchain)

2025

- Python, FastAPI, React, Qwen2.5-VL, Gemini 2.5, Solidity: authenticity verification, price estimation through web scraping, and smart contracts on Ethereum and Sepolia.

Clinical Decision Support (computer vision and RAG)

2025

- Fine-tuned vision-language models (BLIP, CLIP+LLaMa LoRA) and a RAG stack (Meditron-7B, Qdrant, PubMedBERT) to generate medical reports with citations from radiographs.

Health Record, a health tracking platform

2025

- Django, PyTorch: ResNet50 and ResNet18 medical vision across 14 thoracic pathologies, anomaly detection (Isolation Forest, One-Class SVM), a T5 chatbot and a JWT-secured API.

EDUCATION

Engineering Degree, TWIN specialisation (Web, Internet and Network Technologies)

2023 - 2026

ESPRIT, Private Higher School of Engineering and Technology, Tunis

Bachelor's Degree in Business Computing, E-Business

2020 - 2023

ESEN, Higher School of Digital Economy, Manouba

TECHNICAL SKILLS

Languages and frameworks: Python, JavaScript, TypeScript, Java, React, Node.js, Express, FastAPI, Flask, Django, Laravel, PHP

AI and machine learning: LLM Agents, LangGraph, LangChain, RAG, PyTorch, scikit-learn, TensorFlow, YOLO, Computer Vision, spaCy, PaddleOCR

Data and DevOps: MongoDB, PostgreSQL, MySQL, Qdrant, Docker, Git, GitLab CI/CD, SonarQube, AWS, RabbitMQ, Keycloak

Methods: REST APIs, microservices architecture, Scrum, Agile, unit testing, code review

CERTIFICATIONS

CCNA Switching, Routing and Wireless Essentials, Cisco Networking Academy

September 2024

AWS Academy Graduate, Cloud Foundations, AWS Academy (Credly badge)

May 2025

Hashgraph Developer Course, Hedera Hashgraph

October 2025

SDG, Work Ethics and Gender Equity, Honoris United Universities

2024

LANGUAGES

Arabic: native. French: fluent. English: advanced, professional working level.